

Rationally engineered chitin deacetylase from *Arthrobacter* sp. AW19M34-1 with improved catalytic activity toward crystalline chitin

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ARTICLE INFO

Keywords:

Rational design
Chitin deacetylase
Crystalline chitin
Chitosan

ABSTRACT

Chitin and its derivatives have anticoagulant, antimicrobial, and antioxidant properties, but the poor solubility of chitin limits its application in different fields. In this study, site-directed mutagenesis was performed to enhance the deacetylation activity of chitin deacetylases CDA from *Arthrobacter* (ArCE4). The mutant Mut-2-8 with Y172E/E200S/Y201W showed a 2.84-fold and 1.39-fold increase in catalytic efficiency (k_{cat}/K_m) for the deacetylation of (GluNAc)₅ and α-chitin, respectively. These results demonstrated that the mutations significantly improved the activation of ArCE4 on crystalline chitin. The molecular docking study confirmed that the enhancement of catalytic efficiency is due to the extra two hydrogen bonds and one acetyl group. In summary, the activity of Mut-2-8 to insoluble chitin was significantly improved by reactional design, which is beneficial to resolve the issues of the expensive cost of the enzymes and low efficiency. Mut-2-8 exhibits potential applications in the chitosan industry.

1. Introduction

Chitin is a linear polysaccharide of repeating β-1, 4-N-acetyl-D-glucosamine units (GlcNAc). After cellulose, it is the second most ample biopolymer. Chitin is widely found in crustacean shells, fungi, insects, annelids, mollusks, coelenterates, etc. (Machalowski et al., 2019; Tsurkan et al., 2021). Chitin has a highly ordered structure with an excess of crystalline regions. The polymer chains of chitin associate with each other by very strong hydrogen bonding between the amide groups and the carbonyl groups of the nearby chains, resulting in the high insolubility of chitin in the most common solvents (Kardas et al., 2012). Chitin and its derivatives have anticoagulant, antimicrobial, and antioxidant properties, but the poor solubility of chitin limits its application in different fields (Prasasty et al., 2014; Sudatta et al., 2020). Chitosan is derived from chitin by de-N-acetylation, and the degree of deacetylation (DD) should be above 50% (Bai et al., 2020; Bonilla et al., 2019). The protonation of free amino groups makes chitosan soluble in acidic

solutions (Mukherjee et al., 2019; Salama et al., 2020), resulting in broader applications in the food industry, biomedical domain, agriculture, and water treatment (Das et al., 2020; Dong et al., 2020).

At present, commercially available chitosan is mainly produced by traditional chemical methods with a concentrated alkaline solution at high temperature (Borja-Urzola et al., 2020). The chemical methods have several drawbacks like high water footprint, environmental pollution, and heterogeneous products with varying average degrees of polymerization (DP) and fractions of acetylation (FA) (Bhat et al., 2019; Weinhold et al., 2009). Chitin deacetylase (CDA, E.C. 3.5.1.41) belongs to the carbohydrate esterase family 4 (CE4). It catalyzes the removal of acetyl groups from chitin. (Lombard et al., 2014; Tsigos et al., 2000). As a biotechnological alternative to preparing chitosan, the enzymatic deacetylation method has been proposed as a promising tool to produce chitosan owing to its numerous advantages, including eco-friendliness, effortless control, and desired physicochemical properties of the obtained products (Chai et al., 2020; Subramanian et al., 2020).

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<https://doi.org/10.1016/j.carbpol.2021.118637>

Received 1 July 2021; Received in revised form 1 September 2021; Accepted 1 September 2021

Available online 8 September 2021

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Some CDAs have been identified, purified, and characterized from various organisms, including archaea, bacteria, fungi, and insects (Gri-foll-Romero et al., 2018). The CDAs from different sources have distinct catalytic properties, especially the catalytic activity for the de-*N*-acetylation of crystalline chitin (Aragunde et al., 2018). Most CDAs are idle on crystalline chitin, which is the bottleneck that restricts the application of CDAs to prepare chitosan. Several physical and chemical methods have been employed to improve the catalytic activity for the de-*N*-acetylation of crystalline chitin (Gopal et al., 2019; Sebastian et al., 2019). However, if enzymes exhibit high activities on crystalline chitin, they would be beneficial for chitosan preparation.

With the development of enzyme engineering techniques, enzymatic structures have been designed rationally or semi-rationally to improve the catalytic activities and properties (Chen & Arnold, 2020; Zhang et al., 2020). For example, hydrogen bonds around the active site of S-adenosylmethionine synthetase are changed to improve its activity through semi-rational modification (Wang et al., 2019). After rationally engineering the prolyl endopeptidase, its mechanism of enhanced activity toward pathogenic peptides has been investigated by molecular docking (Xiao et al., 2020). In another study, chitinase performs more stable in high temperature after sequence- and structure-based semi-rational design (Xu et al., 2020). Based on transesterification assay combined with virtual docking and iterative saturation mutagenesis, the recognition and activity of *Pseudozyma antarctica* lipase B was improved (Santos et al., 2019).

The CDA from *Arthrobacter* sp. AW19M34-1 (ArCE4) showed low activity toward crystalline chitin such as α -chitin and β -chitin (Tuveng et al., 2017). Engineering ArCE4 to improve its activity toward crystalline chitin could significantly increase their potential applications in an industrial setting. Here we report how rational design aided the improvement of the activity toward crystalline. The mutant showed the highest activity on crystalline chitin. The molecular mechanisms of crystalline chitin activity increase were also investigated.

2. Materials and methods

2.1. Chemicals

α -chitin powder (C7170, average molecular weight 400,000, DD 30%) was purchased from Sigma-Aldrich Co. LLC (USA). Soluble chitosan (average molecular weight 300,000, DD 80%) was purchased from Jiangsu Aoxin Biotechnology Co., Ltd. (China). *N*-acetylchito-oligomers [(GlcNAc)₂₋₆] were purchased from Qingdao BZ Oligo Biotech Co., Ltd. (China). β -chitin was prepared from a squid pen collected from Lianyungang city, Ching, according to Ianiro and Giosia (Ma et al., 2019).

2.2. Plasmids and strains

Ni-NTA His Tag Kit was purchased from Takara (Dalian, China). The pET-21(a) + plasmid, *Escherichia coli* DH5 α , and *E. coli* BL21 (DE3) were purchased from Sangon (Shanghai, China). Strains were grown in LB with 50 μ g/mL of ampicillin (Amp).

2.3. Gene synthesis, expression, and purification

The wild-type ArCE4 gene (GenBank Accession NO. LT630322.1) and mutant gene were synthesized by GenScript Biotech Co. (Nanjing, China). The ArCE4 gene was amplified with the following primers: forward primer, 5'-GGAATTC CATATG CACCACCACCACCAC-CAACCGGAACCGGTGGCGA C-3', reverse primer, 5'-CCGC TCGAG TTACGGGTGGTC TTGAAACGGTG-3' with the synthesized ArCE4 gene as a template. The amplified product was purified and digested with *Nde*I/*Xho*I and then ligated into a *Nde*I/*Xho*I-digested and dephosphorylated pET21a(+) vector. The resulting plasmid named pET-21a(+)-ArCE4 was transformed into *E. coli* DH5 α . Plasmid DNAs were extracted. After gene sequencing, correct plasmids were transformed

into *E. coli* BL21 (DE3). The *E. coli* cells harboring the ArCE4 CDA gene were grown at 37 °C in LB medium containing 50 μ g/mL Amp. Over-expression was induced by adding 0.1 mM isopropyl- β -D-thiogalactopyranoside (IPTG) when OD₆₀₀ reached 0.6–0.8 at 30 °C overnight. After harvesting, the cells were disrupted by sonication in the resuspending buffer [10 mM Tris-HCl buffer with pH 7.0]. The cell debris was removed by centrifugation at 12,000 $\times$ g and 4 °C for 10 min. The supernatant was then rapidly purified by Nickel-nitrilotriacetic acid (Ni-NTA) affinity chromatography (Qiagen, Hilden, Germany). The 4 mL supernatant sample was loaded to the NTA resin column. Following twice washing with 4 mL wash buffer (50 mM NaH₂PO₄, 300 mM NaCl, 20 mM imidazole, pH 8), the protein was eluted twice with 0.5 mL of elution buffer (50 mM NaH₂PO₄, 300 mM NaCl, 200 mM imidazole, pH 8.0). The His-tag elutes were collected and analyzed by 12% SDS-PAGE. The Bradford method determined the protein concentration with bovine serum albumin as the standard (Bradford, 1976).

2.4. Rational design

The three-dimensional structure of AnCDA (PDB ID: 2Y8U) and ArCE4 (PDB ID: 5LFZ) were visualized and aligned with PyMOL (DeLano Scientific LLC, USA). Based on the three-dimensional structure of AnCDA and ArCE4 and structure-based sequence alignment of CE4 deacetylases reported by Tuveng et al. (2017), the amino acid residues are located on the surface of the substrate-binding region excluding conserved catalytic motifs were selected. Each of the amino acid residues defined above was replaced with other 19 standard amino acids, and the homology models of the mutations were created using the Swiss-Model web server (<https://swissmodel.expasy.org/>) with ArCE4 (PDB ID: 5LFZ) as a template (Bordoli et al., 2009). The mutations with increasing negative charges on the surface of the substrate-binding region were selected to perform site-directed mutagenesis. The activities of the mutants with crystal chitin as substrate were determined. The mutant of the first amino acid with the highest activity on crystal chitin was used as the template to perform site-directed mutagenesis for the second amino acid residue. Site-directed mutagenesis for the second amino acid residue was performed until all the selected amino acid residues were mutated. Furthermore, the residues corresponding to three aromatic residues of AnCDA (Y53, Y200, W201) were reported to aid carbohydrate-processing enzymes in substrate binding of the mutant with the highest activity on crystal chitin were mutated into different aromatic residues with site-directed mutagenesis Kit (Table 1).

2.5. Site-directed mutagenesis and screening of mutations with high activity on crystalline chitin

Site-directed mutagenesis was performed using the Takara Muta-nBEST kit (TaKaRa, Dalian, China) with primers list in Supplementary Table 1. After blunting and dephosphorylation, PCR products were digested and inserted into the *Nde*I/*Xho*I site of pET21a (+), and the recombinant plasmids were transformed into *E. coli* DH5 α competent

Table 1
Summary of the specific activity of ArCE4 and mutants.

NO.	Enzymes	Specific activity (U/mg)
–	ArCE4	210.2
Mut-1	K 172 D	313.7
Mut-2	K 172 E	385.8
Mut-2-1	K 172 E/ES 200 FF	446.1
Mut-2-2	K 172 E/ES 200 FW	401.4
Mut-2-3	K 172E/ES 200 FY	468.3
Mut-2-4	K 172E/ES 200 WF	407.1
Mut-2-5	K 172E/ES 200 WW	518.4
Mut-2-6	K 172 E/ES 200 WY	464.8
Mut-2-7	K 172E/ES 200 YF	450.1
Mut-2-8	K 172E/ES 200 YW	539.7
Mut-2-9	K 172E/ES 200 YY	398.4

cells. IPTG induced overexpression, and the recombinant proteins were purified with the method described above. Enzyme activity was measured by a method described below. α -chitin was used as the substrate to screen the mutants. SDS-PAGE assessed the degree of purity on a 12% gel.

2.6. Characterization of recombinant wild-type ArCE4 and Mut-2-8

The optimal pH was measured within a pH range from 3.0 to 10.0 at 30 °C. The residual activity was assayed by incubating the enzyme solution above pH at 40 °C for 2 h to determine the pH stability. The optimal temperature was examined in the temperature range from 25 °C to 60 °C at pH 8.0. The thermostability was determined by assaying the residual activity after incubating the enzyme solution in 20 mM Tris-HCl buffer (pH 7.0) at temperatures ranging from 10 °C to 60 °C for 2 h. Chitin oligomers dimers [(GlcNAc)₂₋₆], α -chitin, and β -chitin were used as substrates to study the substrate specificity of ArCE4 and Mut-2-8. The properties of α -chitin, chitosan, and chitin treated with ArCE4 and Mut-2-8 were observed under the Scanning Electron Microscope (SEM-JSM-6390LA, Japan). The enzyme solution of 50 mL was mixed with 50mMphosphate buffer with pH 7.0 at 30 °C for 48 h. The chitin treated with enzyme solution was centrifuged for 5000 $\times$ g in 15 min and dried to a constant weight at 100 °C. The sample was coated on the double-sided adhesive tape pasted on the metallic stub subjected to gold-coated 100A in SEM at 20 kV. Alpha-chitin, α -chitin after enzymatic hydrolysis, and commercial chitosan were also characterized by Fourier transform infrared spectroscopy (FTIR). The absorbance ratio of A1655/A3450 was investigated for the determination of the deacetylation degree.

The kinetic parameters for the wild-type ArCE4 and the mutant were calculated by using various concentrations of (GluNAc)₅ (2, 4, 6, 8, 10 mM) and α -chitin suspension (0.02%, 0.04%, 0.06%, 0.08%, 0.1%, W/V) with standard assay, separately. Kcat was calculated using $1/V = K_m/V_{max} \times 1/S + 1/V_{max}$, where E was the molar concentration of enzymes and the unit was μ M. Km and Vmax were determined from Lineweaver–Burk plots using Microsoft Excel.

The properties of α -chitin, water-soluble chitosan, and chitin treated with wild-type ArCE4 and Mut-2-8 were observed under the Scanning Electron Microscope (SEM-JSM-6390LA, Japan). The purified recombinant wild-type ArCE4 and Mut-2-8 were added to 50 mM phosphate buffer with pH 7.0 to 100 mL. The reaction mixture consisted of 0.1 g chitin powder, and the total concentration of protein present in each reaction was constant (0.2 mg/mL). The procedures were performed at 140 rpm, 35 °C for 72 h. The chitin treated with enzyme solution was centrifuged for 5000 $\times$ g in 15 min and dried to a constant weight at 100 °C. The sample was coated on the adhesive double-sided tape pasted on the metallic stub subjected to gold-coated 100A in SEM at 20 kV.

2.7. Determination of the degree of deacetylation (DD)

FT-IR spectra of the samples of α -chitin, water-soluble chitosan, and chitin treated with wild-type ArCE4 and Mut-2-8 were measured through FT-IR spectra (Nicolet-iS10) at the wavenumber region from 400 to 4000 cm^{-1} at an interval of 2 cm^{-1} . The samples were freeze-dried before analysis. The absorbance at 1655 cm^{-1} (amide I) and 3450 cm^{-1} (OH group) and used to calculate the DD as the following eq.

$$DD(\%) = 1 - \frac{A_{1655} - A_{3450}}{1.33}$$

2.8. Enzymatic activity determination

The reaction mixtures of 5 mL, containing 1 μ M purified ArCE4, 2 mM or 5 mg/mL substrate and 10 μ M CoCl₂ in 25 mM Tris-HCl buffer (pH 8.0), were shaken and incubated at 37 °C for 24 h. Reactions were quenched by adding acetonitrile to mixtures at a final concentration of

50% (V/V), and acetate release was measured using the K-ACETRM Kit (Megazyme, Ireland). All experiments were repeated 3 times and corrected for background from enzyme-free control reactions. One unit of enzymatic activity was defined as the amount of enzyme liberating 1 μ M acetic acid per minute.

2.9. Prediction and analysis of the 3D structure of mutant ArCE4

The sequences of mutant ArCE4 were submitted to SWISS-MODEL to construct its 3D structure using 5LFZ as the template. PyMOL visualized the resulting pdf files. The surface charges of the mutants were generated using PyMOL (Action > generate > vacuum electrostatics > protein contact potential) (<https://pymol.org/2/>).

2.10. Molecular docking of (GlcNAc)₂ complexes with Mut-2-8

The ArCE4 protein structure (PDB ID: 5LGC) was retrieved from the RCSB Protein Data Bank and prepared by removing all water molecules, assignment of Gasteiger partial charges, and adding polar hydrogen. Ligand position in ArCE4 protein was set at grid coordinates x = −22.4, y = 20.3 and z = 32.5. Genetic algorithm (GA) parameters were assigned at 100 GA run and population size of 150. Molecular docking calculations were performed by using Autodock 4.2 with the aid of Auto Dock Tools 1.5.6 (Forli et al., 2016). Autodock Vina 1.1.2 was used to conduct semi-flexible docking between mutant ArCE4 and (GlcNAc)₂. Determine the coordinates and box size of Vina molecular docking. To increase the accuracy of calculation, the parameter exhaustiveness is set to 20. The result with the least affinity was selected for subsequent analysis, such as hydrogen bond calculations. Discovery Studio v3.1 and PyMOL v2.0 were utilized to visualize the ligand-receptor complexes.

3. Results and discussion

3.1. Rational design

The optimal pH of ArCE4 is pH 7.0. Based on the three-dimensional structure alignment of AnCDA and ArCE4, and the surface charge distributions of ArCE4 analysis (Supplementary Fig. 1), the residues with positive charges at pH 7.0 near the substrate-binding region surface are 146 Y, 171 W, and 172 K. The amino acid sequences alignment reported by Tueng et al. (2017) showed that 146 Y located in conserved catalytic motif MT3 and involved in catalysis (Marked with pink asterisks), 171 W located in conserved catalytic motif MT4, and therefore only 172 K was selected to be replaced with other 19 standard amino acids. The three-dimensional structure of the corresponding mutants was predicted, and their surface charge distributions were calculated. Compared with ArCE4, K 172 D and K 172 E have more negative charges on the surface of their substrate-binding regions (Supplementary Fig. 1). The sequence alignment and 3D structure analysis showed that Y 61, E 200, and S 201 of ArCE4 corresponded to the aromatic residues near the catalytic center of AnCDA (Y 53, Y 200, W 201). E 200 and S 201 were mutated into different aromatic residues combinations (Table 1).

The chitin deacetylase from *Aspergillus nidulans* FGSC A4 (AnCDA) acts on crystalline chitin at a meager rate (Liu et al., 2017). Liu et al. analyzed the structure of AnCDA. They found that a surplus of negative charge on the surface of the substrate-binding region that could interact with the non-reducing end of a more extended oligomeric substrate and a cluster of significant aromatic residues on the surface of the protein near the catalytic center (Y53, Y200, W201) and may aid AnCDA in binding longer crystalline chitoooligomers (Liu et al., 2017).

3.2. Site-directed mutagenesis and screening of mutations with high activity on crystalline chitin

Based on the analysis results described above, K172 was selected to be replaced with D and E, which have more negative charges on the

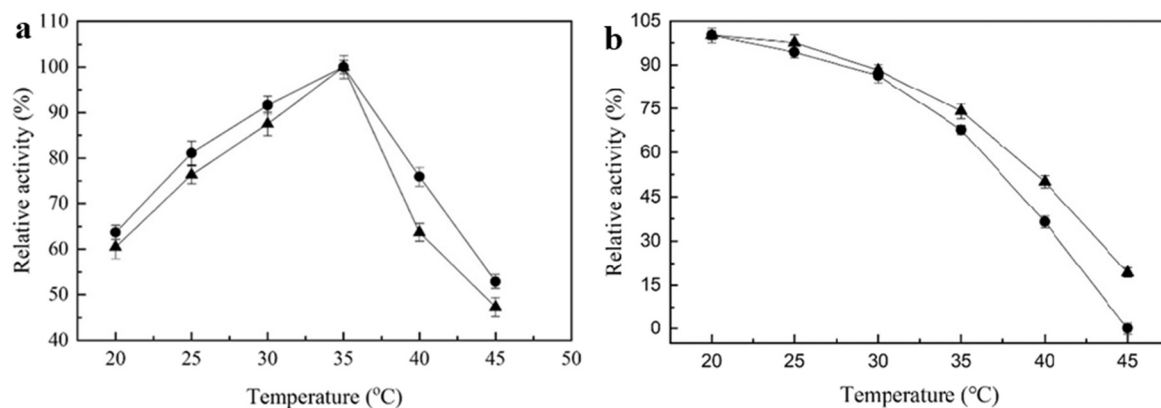


Fig. 1. Effect of temperature on the activity (a) and stability(b) of ArCE4 (▲) and Mut-2-8 (●).

surface of their substrate-binding regions (Fig. 1). Then, the Glu 200 and Ser 201 of mutant K172E were introduced to perform site-directed mutagenesis to different aromatic residues combinations. The wild-type and mutant ArCE4 were purified (Supplementary Fig. 2). The activity of different mutants to crystalline chitin (α -chitin) was determined. Interestingly, all the mutants showed higher specific activity to α -chitin. The mutated enzymes K172D and K172E displayed higher specific activity to α -chitin by increasing to 313.7 U/mg and 385.8 U/mg, respectively (Table 1). The activity of Mut-2-8 to α -chitin increased up to 539.7 U/mg, which was 2.57-fold that of the wild enzyme, a remarkable improvement in activity to α -chitin (Table 1).

These results demonstrated that the mutations significantly improved the activation of ArCE4 on crystalline chitin. Currently, commercially available chitosan is still mainly produced by traditional chemical methods with many drawbacks. CDA catalyzes the conversion of chitin to chitosan and represents a more environmentally friendly and easier to control alternative to overcome the disadvantages of chemical methods. However, several CDAs are idle on crystalline chitin, which restricts the application of CDAs to prepare chitosan. To better meet the industrial application requirement, the activity of CDAs on crystalline chitin should be improved.

3.3. Characterization of recombinant wild-type ArCE4 and Mut-2-8

The effect of temperature on the activity and stability of wild-type and Mut-2-8 were determined with α -chitin as substrate at various temperatures ranging from 20 to 45 °C. Both wild-type ArCE4 and Mut-2-8 showed the highest activity at approximately 35 °C (Fig. 1a). The residual enzymatic activities of the wild type and Mut-2-8 at temperatures ranging from 10 °C to 60 °C for 2 h are shown in Fig. 1b. And the Mut-2-8 exhibited higher residual activity compared to the wild enzyme,

meaning better thermostability. Thermostability is considered an essential property for utilizing the enzyme in chitosan preparation. The pH dependence of wild-type and Mut-2-8 were examined in various buffers at pH values that ranged from 3.0 to 10.0 under standard assay conditions. The results showed that the optimal pH of wild-type and Mut-2-8 were 7.0 (Fig. 2a). The pH stability curve (Fig. 2b) showed that wild-type and Mut-2-8 are stable over a broad pH range of 5.0–9.0 with more than 60% residual activity after incubated at 40 °C for 2 h. (GlcNAc)₅ is the best substrates for the wild type and Mut-2-8

The Michaelis constants (K_m values) of wild type ArCE4 and Mut-2-8

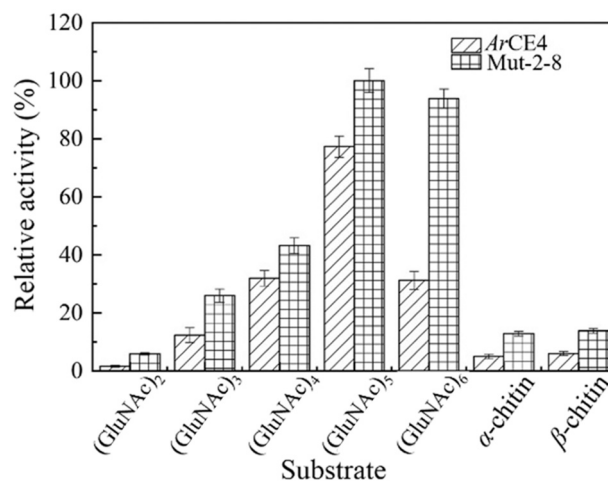


Fig. 3. Effect of substrates on the activity of ArCE4 and Mut-2-8.

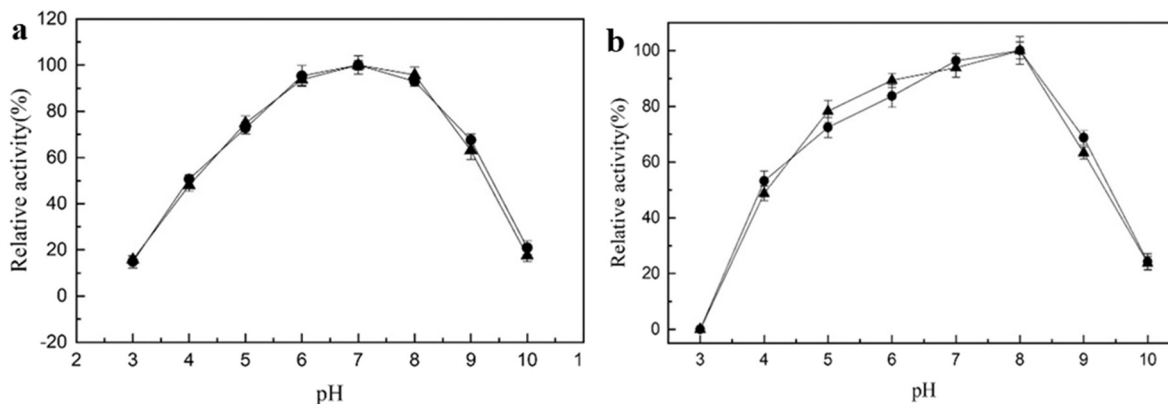


Fig. 2. Effect of pH on the activity (a) and stability(b) of ArCE4 (▲) and Mut-2-8 (●).

to the (GlcNAc)₅ and α -chitin are given in Table 2. Mut-2-8 showed significantly decreased turnover rates (kcat), increased substrate binding affinities (decreased Km), and kcat/Km to both (GlcNAc)₅ and α -chitin as substrate. The Km value of the Mut-2-8 for the acetyl groups of (GlcNAc)₅ is 1.95 mM (i.e., 0.39 mM (GlcNAc)₅). Km value of several CDAs for chito-oligosaccharides have been reported, such as the CDA from *Mucor rouxii* (MrCDA) toward (GlcNAc)₆ (1.3 mM) (Martinou et al., 1995), the AnCDA for (GlcNAc)₆ (0.072 mM) (Liu et al., 2017), and the CDA from *Colletotrichum lindemuthianum* (ClCDA) for (GlcNAc)₆ (0.048 mM) (Hekmat et al., 2003). DMSO strengthens chitin deacetylase-chitin interaction, improving the catalytic activity of chitin deacetylase from *Cryptococcus laurentii* on ethylene glycol chitosan, the Km is 0.53 mM. There was no report on the Km value of CDA to α -chitin because most CDAs showed low activity toward crystalline chitin. Although the Km value of the Mut-2-8 for (GlcNAc)₅ is higher than those of AnCDA and ClCDA, the activity of Mut-2-8 to crystalline chitin significantly improved by reactional design made its potential applications in the chitosan industry.

The magnified image from the scanning electron microscopy (SEM) of α -chitin, water-soluble chitosan, ArCE4 treated α -chitin, and Mut-2-8 treated α -chitin were presented in Fig. 4.

Chitin before and following CDA treatment was observed to have distinct changes in the surface architecture. Some distinctly arranged crystalline microfibrils were visible at the surface of commercial α -chitin (Fig. 6a) but disappeared at the surface of water-soluble chitosan, and the surface of it was looser and more porous (Fig. 6b). CDA treatment decreased the characteristic microfibrillar structure at the surface of chitin and made it looser and more porous. The characteristic microfibrillar structure is still visible in both ArCE4 treated and Mut-2-8 treated α -chitin. Still, the level of characteristic microfibrillar structure of Mut-2-8 treated α -chitin is somewhat lower than that of ArCE4 treated α -chitin (Fig. 6c) (Fig. 6d). Pareek et al. (2013) reported that the commercial chitin has a compact structure with some distinctly arranged crystalline microfibrils visible at its surface. Similarly, after deacetylation by CDA from *Nitratireductor aquimarinus*, the surface of the chitin became loose, porosity, and many crystal fibers disappeared (Chai et al., 2020).

0.5 mg/mL of α -chitin were incubated 5 μ g ArCE4 or Mut-2-8 at 30 °C for 72 h, and then the deacetylation degree of the α -chitin was determined by FT-IR to use the method described previously (Chai et al., 2020). All original freeze-dried chitin and CDA-treated chitin samples were analyzed by FT-IR (Fig. 5). The degree of deacetylation of all the samples was evaluated from the FT-IR spectra using the absorption ratio A1560/A1030 with the equation described above. The degree of deacetylation of α -chitin, water-soluble chitosan, and α -chitin treated with wild-type ArCE4 and Mut-2-8 were 31.35%, 84.77%, 44.87%, and 59.48%. The DDs of the untreated chitin treated with AnCDA, MrCDA, CDA from *Nitratireductor aquimarinus* (NaCDA) was about 50%, less than 0.5%, and 56.26%, respectively Chai et al., 2020; Liu et al., 2017; Martinou et al., 1995).

Converting chitin into chitosan with chitin deacetylases has shown advantages because the reaction is mild and environment-friendly compared to the chemical method. However, enzymatic methods have some significant drawbacks. The biggest is the high cost because CDAs used for deacetylation are considerably more expensive than the generic

bases used in chemical methods. Moreover, the reaction efficiency of CDA is low, let alone that most reported CDAs have no activity or low activity on crystalline chitin (Bonin et al., 2020; Kou et al., 2021; Schmitz et al., 2019). CDA activity on crystalline chitin was enhanced by oxidative cleavage of the surface polymer chains by lytic polysaccharide monooxygenases (Liu et al., 2017). The activity of Mut-2-8 to crystalline chitin was significantly improved by reactional design, which is beneficial to resolve the issues of the expensive cost of the enzymes and low efficiency.

3.4. Prediction and analysis of the three-dimensional structure of mutant ArCE4

Mut-2-8 showed significantly decreased kcat, Km, and kcat/Km to both chitinpentase and α -chitin as substrate. The 3D structure of Mut-2-8 was predicted with SWISS-MODEL using 5LFZ as the template. The surface charges of Mut-2-8 were analyzed and visualized with PyMOL (Fig. 6). Mut-2-8 has an excessive negative charge in the substrate-binding region that interacts with the more extended substrate, leading to enhanced substrate accessibility to chitin. A similar result was found in the chitin deacetylase from *Saccharomyces cerevisiae* (ScCDA₂) (Zhu et al., 2019) and AnCDA (Liu et al., 2017).

3.5. Molecular docking of (GlcNAc)₂ complexes with Mut-2-8

To investigate the reason why Mut-2-8 showed the increased activity for chitinpentase. (GlcNAc)₂ was inserted Mut-2-8 model, which was predicted based on the crystal structure 5LFZ. Molecular docking was performed on the complexes to identify amino acids that interact with substrates. Compared to the structure of ArCE4 in complex with (GluNAc)₂ reported by Liu et al. (2017), which has one acetyl group bind to the catalytic region, the structure of Mut-2-8 in complex with (GluNAc)₂ have two acetyl groups bind to the catalytic region (Fig. 7a,b). Furthermore, there are six hydrogen bonds between Mut-2-8 and (GluNAc)₂, while only four hydrogen bonds between ArCE4 and (GluNAc)₂ (Fig. 7c). The extra two hydrogen bonds and one acetyl group may reinforce the binding of (GluNAc)₂ and Mut-2-8. This is maybe the reason why Mut-2-8 showed higher catalytic efficiency for chitinpentase than that of ArCE4 (Fig. 8). Xiang et al. (2019) reported that the new hydrogen bonds of the mutant of xylanase might improve catalytic efficiency.

The CDAs have a highly conserved α/β barrel structure, while the substrate-binding cleft composing of six loops (Loop 1 to Loop 6) located on one side of the barrel was quite diversified in structure. VcCDA has long loops shaping a deep binding cleft and prefers a short substrate. While ArCE4 has short loops 1 to 6, resulting in an open and shallow cleft and allowing binding of longer substrates on a shallow binding cleft, shows increasing activity with longer substrates (Aragunde et al., 2018). In the present study, the extra hydrogen bonds and acetyl group reinforce the enzyme-substrate interactions and may lead to Mut-2-8 showed higher catalytic efficiency for (GluNAc)₅ than that of ArCE4. Moreover, although the chitin binding motif is unchanged, the open and shallow cleft of Mut-2-8 allowing it binding of longer substrates, both the improved negative charges on the surface of the substrate-binding region and the aromatic residue in substrate binding may improve reinforce the Mut-2-8 and substrate interactions, and lead to the higher activity on insoluble chitin. Similarly, the carbohydrate-binding modules (CBM) enhanced the deacetylase activity of some CDAs on insoluble chitin through increasing enzyme-substrate interactions (Hoßbach et al., 2018).

4. Conclusions

In this research, we first collect the characteristics of CDAs active to crystalline chitin and rationally design ArCE4 with site-directed mutation. The amount of acetic acid released in the enzymatic reaction is

Table 2

Kinetic parameters of wild-type and Mut-2-8 using α -chitin and (GlcNAc)₅ as substrate.

Enzyme	Substrates	Km (mM) ^a	K _{cat} (min ⁻¹)	K _{cat} /Km
ArCE4	(GlcNAc) ₅	3.66	23.01	6.28
	α -chitin	7.59	5.77	9.77
Mut-2-8	(GlcNAc) ₅	1.95	34.8	17.85
	α -chitin	5.73	9.93	13.61

^a The concentration of acetyl groups.

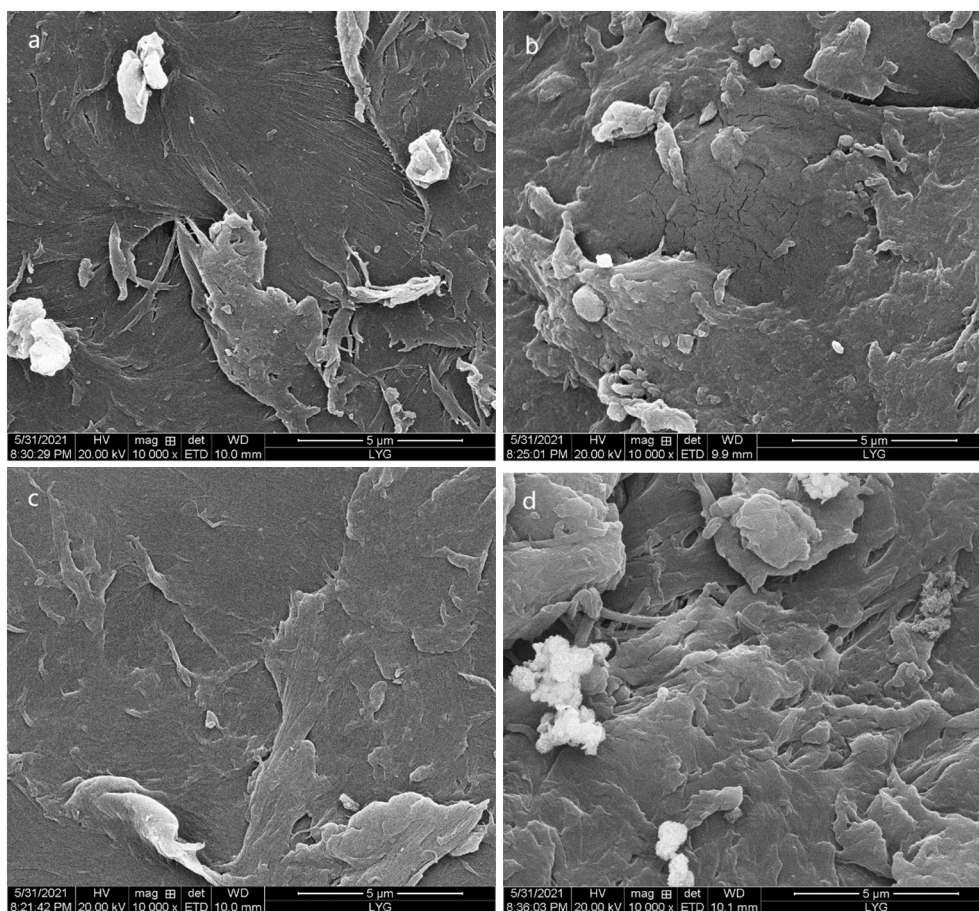


Fig. 4. SEM images of α -chitin (a), water-soluble chitosan (b), ArCE4 treated α -chitin(c), and Mut-2-8 treated α -chitin(d).

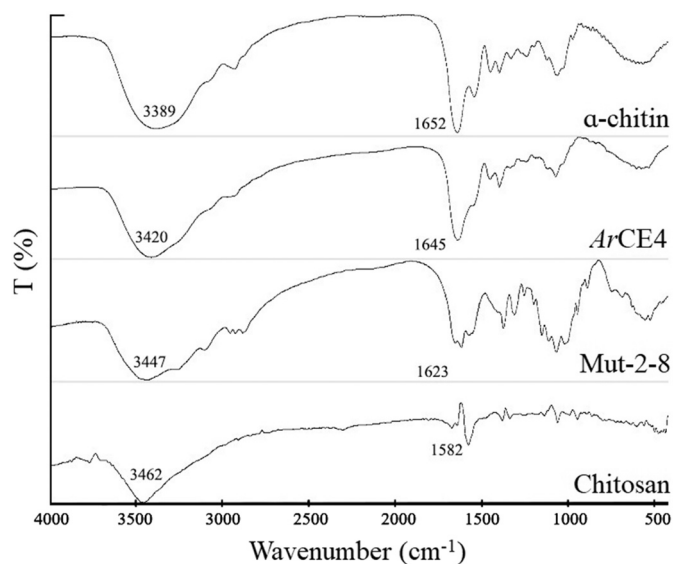


Fig. 5. FT-IR spectra of α -chitin, chitosan, α -chitin treated by ArCE4 or Mut-2-8.

detected by IC to screen the mutant enzyme with increased activity. After reacting with Mut-2-8, α -chitin releases a total of 10.18 μ M acetic acid, and its deacetylation degree reaches 8.13%, which is 2.57 times that with ArCE4. In addition, the catalytic activities of Mut-2-8 on chitin oligomers are also improved. Comparing the structures after docking

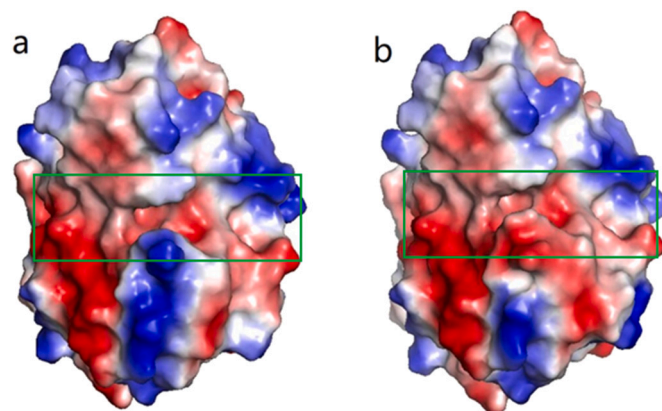


Fig. 6. The charge distributions on the surface of the substrate-binding region (The green boxes) of ArCE4 (a) and Mut-2-8 (b). Note: Red represents negative charges, and blue represents positive charges.

with (GluNAc)₂, it was found that the binding of Mut-2-8 to acetyl groups is more stable. Exposed aromatic residues have also been confirmed to increase catalytic activity. After the mutation, the enzymatic properties don't change much, but the temperature stability is slightly improved, and the optimal condition for deacetylation is neutral.

CRediT authorship contribution statement

Zhiwen Ding: Investigation, Data curation, Writing – original draft,

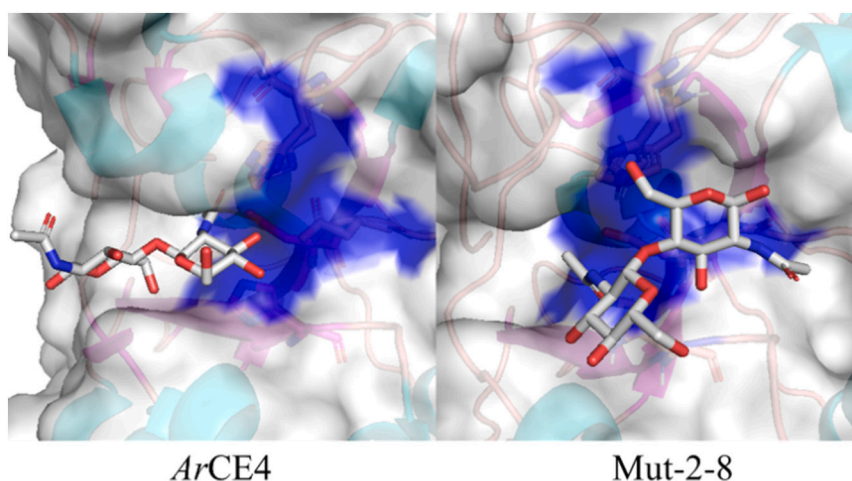


Fig. 7. The catalytic region bound to (GluNAc)₂. The surface of the catalytic region was colored blue, and (GluNAc)₂ was showed as white sticks. Note: The surface of the catalytic region was colored blue, and (GluNAc)₂ was showed as white sticks.

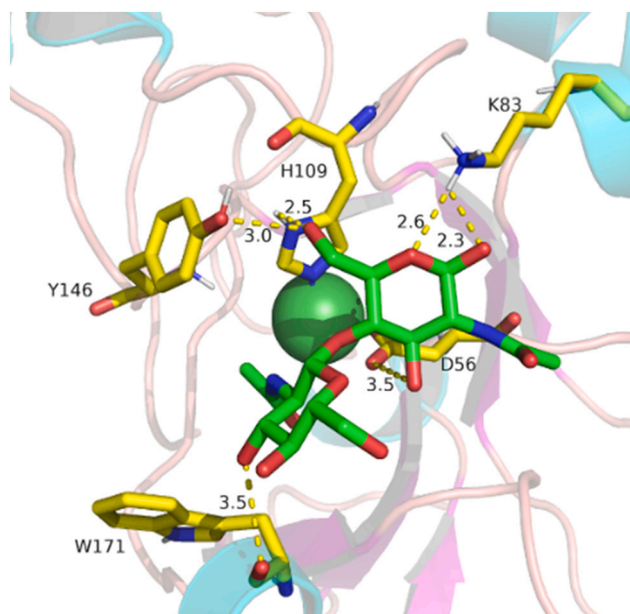


Fig. 8. The ligand-bound in subsites between Mut-2-8 and (GluNAc)₂.

Writing – review & editing. **Sibtain Ahmed:** Methodology, Software. **Jiahao Hang:** Data curation. **Haoyu Mi:** Data curation, Formal analysis. **Xiaoyue Hou:** Data curation. **Guang Yang:** Validation. **Zhifa Huang:** Resources. **Xiaoyue Lu:** Conceptualization, Investigation, Visualization. **Yaowei Fang:** Funding acquisition, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare no conflict of interest.

Acknowledgment

This work was supported by the National Natural Science Foundation of China (31772016), Jiangsu Province Marine Science and Technology Innovation Project (JSZRHYKJ202008, HY2018-10), the Priority Academic Program Development of Jiangsu Higher Education Institutions, The Key Natural Science Foundation of the Jiangsu Higher Education Institutions of China (20KJA550001), the National Natural Science

Foundation of Jiangsu Ocean University (KQ20041), Project“333” of Jiangsu Province, Open-end Funds of Jiangsu Key Laboratory of Marine Bioresources and Environment (SH20191204), Jangsu Province College Students' innovation and entrepreneurship training program (SZ202111641638001), and “521 high-level talents training project” in Lianyungang.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.carbpol.2021.118637>.

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